



CLOTHING STORE RETAIL TECHNOLOGY

Hardware Deployment & Shop Computer Setup Manual

A step-by-step engineering and operations guide for rolling out D'RROW POS on physical store computers, thermal receipt printers, USB barcode scanners, and clothing tag printers.

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Target Environment: Windows 10/11 & Linux LTS

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Retail Hardware & Shop Computer Deployment

This manual covers the physical installation, peripheral wiring, operating system setup, database networking, and peripheral driver configuration needed to deploy the **D'RROW Clothing Store POS and Inventory Management System** in a commercial apparel retail environment.

1. Peripheral Hardware Specifications & Compatibility

The D'RROW POS software is built using pure Java standards and standard operating system abstraction layers. It works out-of-the-box with standard retail peripherals:

Hardware Device	Recommended Specifications	Standard Retail Models
POS Counter PC	Intel Core i3/i5 or AMD Ryzen, 8 GB RAM, 128 GB SSD, Windows 10/11 64-bit or Ubuntu 22.04 LTS	Dell OptiPlex, HP ProDesk, Posiflex, All-in-One Touch PC
Thermal Bill Printer	80mm direct thermal line printer, auto-cutter, USB + Ethernet interface, ESC/POS compatible	Epson TM-T20III / TM-T88VI, Xprinter XP-N160M, Star Micronics TSP143
Barcode Scanner	Handheld 1D/2D Laser/CCD Scanner, USB HID Keyboard Wedge mode, auto-scan stand	Honeywell Voyager 1200g, Zebra Symbol LS2208, Datalogic QuickScan
Clothing Tag Printer	Thermal transfer or direct thermal barcode label printer (203 DPI, 50mm width)	Xprinter XP-365B, Zebra ZD220, TSC TE200, or A4 Laser Printer (21 tags/sheet)
Electronic Cash Drawer	Heavy-duty steel cash drawer, RJ11/RJ12 connector, 24V solenoid pulse kick	Maken MK-410, Posiflex CR-4000, APG Series 100

Counter Layout & Electrical Grounding Guidelines

Retail checkout counters require strict electrical hygiene. Electrostatic discharge (ESD) caused by synthetic fabrics and foot friction can reset USB ports or freeze printer control boards if not properly grounded:

- **Uninterruptible Power Supply (UPS):** Connect the counter PC, receipt printer, and network switch to a dedicated online or line-interactive 650VA+ UPS. Sudden power cuts in retail environments corrupt active database transaction logs if writing mid-commit.
- **Direct Rear Motherboard Ports:** Always connect barcode scanners and thermal printers directly to the rear USB 3.0/2.0 ports of the PC. Do not use unpowered plastic USB extension hubs, as they frequently drop voltage when scanners pull peak current during laser firing.
- **Cable Conduit Separation:** Keep 240V AC power adapters physically separated from the scanner and cash drawer signal cables by at least 10 cm to eliminate electromagnetic interference (EMI).



Figure 1: High-level System and Physical Hardware Architecture

2. Single-Counter vs. Multi-Counter Network Topologies

Depending on the size of the clothing boutique, D'RROW POS supports two standard deployment topologies:

Topology A: Single-Counter Standalone Setup (Small Boutique)

In a single-counter setup, one computer serves both as the **Database Server** (running MySQL 8.x) and as the **Cashier Checkout Terminal** (running the D'RROW Swing Application). All peripherals (barcode scanner, 80mm thermal receipt printer, tag printer, and cash drawer) connect directly to this machine via USB ports.

- **Configuration:** `db.host=localhost` in `config/db.properties`.
- **Advantage:** Minimal hardware cost, zero network cabling, zero latency.

Topology B: Multi-Counter Local Network Setup (Department Stores)

For stores with multiple billing counters and a stock warehouse office:

- **Back-Office Server PC:** Dedicated machine hosting MySQL 8.x and automated hourly backups. It has a static local IP address (e.g., `192.168.1.100`).
- **Counter 1 (Cashier PC):** POS Terminal + USB Scanner + Thermal Receipt Printer. Configured with `db.host=192.168.1.100`.
- **Counter 2 (Cashier PC):** POS Terminal + USB Scanner + Thermal Receipt Printer. Configured with `db.host=192.168.1.100`.
- **Warehouse Station:** Inventory Management + Stock Inward Receiving + Barcode Clothing Tag Printer. Configured with `db.host=192.168.1.100`.

3. MySQL Server Network Hardening for Multi-Counter Stores

To allow cashier computers to communicate with the central database server across the store's local Wi-Fi or Ethernet switch, configure MySQL as follows:

Step 1: Enable Network Listening in `my.ini` / `my.cnf`

Open your MySQL configuration file (located at `C:\ProgramData\MySQL\MySQL Server 8.0\my.ini` on Windows, or `/etc/mysql/my.cnf` on Linux):

```
# Allow connections from any IP on the store LAN
bind-address = 0.0.0.0
port = 3306
```

Step 2: Create a Dedicated POS Network User

Open MySQL Command Line Client on the Server PC and grant remote access to the store subnet:

```
CREATE USER 'drrow_cashier'@'192.168.1.%' IDENTIFIED BY 'DrrowSecure2026!';
GRANT ALL PRIVILEGES ON drrow_pos.* TO 'drrow_cashier'@'192.168.1.%';
FLUSH PRIVILEGES;
```

Step 3: Open Windows Firewall Port 3306

On the Server PC, open Windows Defender Firewall with Advanced Security:

1. Click **Inbound Rules** → **New Rule...**
2. Select **Port** → **TCP** → Specific local ports: 3306.
3. Select **Allow the connection** → Apply to **Domain** and **Private**.
4. Name the rule: MySQL D'RRROW POS Server.

4. Handheld USB Barcode Scanner Installation

Standard retail scanners (Honeywell, Zebra, Datalogic) operate in **USB HID Keyboard Wedge** mode. This means the scanner acts as an ultra-fast automated typist.

Optical Resolution & Apparel Scanning Geometry

Clothing tags present unique scanning challenges due to fabric flex, glossy plastic laminates, and security hangtag strings. To guarantee 100% first-pass read rates:

- **Presentation Mode / Hands-Free Stand:** Place the scanner in its gooseneck counter stand with auto-sensing enabled. Cashiers swipe the clothing tag across the beam with two hands without needing to lift the gun.
- **Symbology Prioritization:** Disable unnecessary symbologies (such as UPC-E, Codabar, MSI) in the scanner firmware settings. Retain only **Code 128** and **EAN-13**. This reduces decode latency from 120 ms down to under 25 ms per garment.



Figure 2: Real-time Barcode Scanning and Atomic Checkout Execution

Configuration Rule: Carriage Return (Enter) Suffix

The D'RROW POS terminal expects the scanner to send an [ENTER] key immediately after the barcode digits. This instantly triggers the database query and adds the clothing variant to the cart.

- Scan the **"Reset to Factory Defaults"** barcode in the scanner manufacturer's manual.
- Scan the **"Add CR / Enter Suffix"** configuration barcode.
- **Test Procedure:** Open Windows Notepad. Scan any clothing tag. The barcode number (e.g. DRTS000002) should appear, and the blinking cursor must automatically jump to the next line.

5. 80mm Thermal Receipt Printer & Cash Drawer Integration

Thermal printers use standard ESC/POS command protocol and print on 80mm (or 58mm) thermal paper rolls without ink or ribbons.

Physical Cabling:

1. Connect the printer's 24V power adapter to the wall outlet.
2. Connect the printer's USB cable to the POS computer.
3. Connect the electronic cash drawer's RJ11/RJ12 cable directly into the **DK (Drawer Kick)** port at the back of the thermal printer.

Windows Printer Driver Setup:

1. Install the manufacturer's official Windows driver (e.g., Epson Advanced Printer Driver or Xprinter Driver).
2. Go to Windows **Settings** → **Bluetooth & devices** → **Printers & scanners**.
3. Select your 80mm printer and click **Set as default**.
4. Under **Device Settings**, verify paper size is set to **80mm x Receipt (or 80 × 297mm)**.
5. Under **Cash Drawer** properties, set **"Open Drawer #1 Before Printing"** or **"After Printing"**. Now, whenever a bill is completed in D'RROW POS, the printer fires a 24V pulse and the cash drawer automatically pops

open!

6. Barcode Clothing Tag Label Printer Setup

Attaching durable price tags to garments requires clear, legible Code 128 barcodes that scan reliably without reflection or fading:

Method A: Dedicated 50×30mm Roll Label Printer (Recommended)

- Load a roll of 50mm width × 30mm height jewelry/clothing sticker labels.
- Adjust the roll guide arms so the paper feeds straight without skew.
- Perform a **Media Gap Sensor Calibration** (hold the FEED button until the light flashes blue/purple).
- In D'RRROW POS, open the **Barcode Tags** tab, choose "*Roll Label (50×30mm)*", and click Print.

Method B: Standard A4 Sticker Sheets (Office Laser/Inkjet)

- Use standard 3-column × 7-row A4 pre-cut sticker sheets (21 labels per page).
- In D'RRROW POS, choose "*A4 Sheet (21 Tags/Sheet)*". The system automatically tiles the tags into a uniform grid.

7. Windows Kiosk Mode & Auto-Start Script

To ensure the POS application launches automatically whenever the store computer is turned on in the morning, create a dedicated batch startup script:

Create run_drrow_pos.bat in the project folder:

```
@echo off
title D'RRROW Clothing Store POS Terminal
cd /d "C:\DRROW\drrow-pos"

REM Wait 3 seconds for local MySQL service to start
timeout /t 3 /nobreak >nul

REM Launch D'RRROW POS with optimized memory settings
start javaw -Xms256m -Xmx1024m -jar target\drrow-clothing-pos-1.0.0.jar
exit
```

Install to Windows Startup:

1. Press win + R, type shell:startup, and press Enter.
2. Create a shortcut to run_drrow_pos.bat in this Startup folder.
3. Now, whenever staff power on the counter PC in the morning, D'RRROW POS opens straight to the login screen!

8. Automated Daily Database Backup Script

Retail store data must never be lost. Configure an automated daily backup scheduled every evening after store closing:

Create daily_backup.bat:

```
@echo off
set BACKUP_DIR=C:\DRROW_Backups
set TIMESTAMP=%date:~10,4%%date:~4,2%%date:~7,2%_time:~0,2%%time:~3,2%
set TIMESTAMP=%TIMESTAMP: =0%

if not exist "%BACKUP_DIR%" mkdir "%BACKUP_DIR%"

echo Starting D'RRROW POS Database Backup...
"C:\Program Files\MySQL\MySQL Server 8.0\bin\mysqldump.exe" -u root -proot123 --routines --triggers drrow_pos > "%BACKUP_DIR%\drrow_pos_backup_%TIMESTAMP%.sql"

echo Backup completed successfully at %BACKUP_DIR%\drrow_pos_backup_%TIMESTAMP%.sql
```

Automate via Windows Task Scheduler:

- Open **Task Scheduler** → **Create Basic Task**.
- Trigger: **Daily at 10:30 PM**.
- Action: **Start a Program** → Select C:\DRROW\daily_backup.bat.
- Keep an external USB flash drive plugged into the PC and mirror the C:\DRROW_Backups folder weekly for total disaster recovery.

9. Hardware Troubleshooting & Rapid Diagnostics

Symptom	Probable Cause	Resolution Procedure
Scanner beeps but doesn't add item to cart	The POS barcode input field does not have focus, or scanner lacks Enter suffix.	Click inside the gold SCAN BARCODE box. Verify scanner sends Enter by testing in Windows Notepad.
Printer prints blank paper or garbled symbols	Paper roll inserted upside down (thermal coating is only on one side).	Flip the paper roll around so the glossy surface faces the thermal print head. Perform printer self-test.
Cash drawer does not kick open	RJ11 cable is loose or plugged into wrong port; driver pulse disabled.	Verify RJ11 cable is plugged into printer's DK port (not phone jack). Enable "Open Drawer" in Windows printer properties.
Counter PC shows "Database Connection Failed"	Server PC is powered off, server IP changed, or firewall blocking port 3306.	Verify Server PC is on. Ping server IP (e.g. ping 192.168.1.100). Ensure port 3306 is open in Windows Firewall.
Barcode tags skip labels or print off-center	Label gap sensor is miscalibrated or dust covering optical sensor.	Wipe sensor with isopropyl alcohol swab. Hold printer FEED button to recalibrate label gap detection.